

AHMED NABIL

Full Stack Engineer

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Experience

SKMD, Software Engineer Intern 

Sept 2024 – Present

- **Implemented a migration from Material-UI (MUI) v4 to v5** across **265 files**, including critical components like the Grid system and DatePickers.
- **Standardized date management** by migrating the codebase from **Moment.js** and **date-fns** to **Day.js** across **36** core date files, reducing the application bundle size by approximately **65kB**.
- **Architected scalable cloud integrations**, designing a Firebase and Firestore image upload pipeline alongside an OAuth2-secured Google Sheets data export system.

Ebdaa Bahrain, Freelance Developer 

Jul 2025 – Aug 2025

- Refactored a single-page PHP application into **15+** modular components and optimized library loading, improving Lighthouse performance from **38 to 92 (+142%)**.
- Reduced First Contentful Paint from **7.1s to 0.6s (91% faster)** and Total Blocking Time from **860ms to 10ms (98% reduction)**.

Skills

- Self-taught backend and frontend development through independent study and hands-on projects.
- Led and coordinated university project teams, successfully managing tasks and resolving development challenges.
- Managed time effectively to learn multiple technologies and programming languages simultaneously.

Education

Capital University (Previously Helwan), B.Sc. in Computer Science and Artificial Intelligence

Sept 2022 – Jun 2026

- **Relevant Coursework:** Data Structures, Algorithms, Software Engineering, Information Technology, Database Systems, Operating Systems.

Projects

Real-Time Watch Together Platform (ModdieView , ModdieView Backend )

A real-time, synchronized media viewing application to watch videos together with friends.

- Implemented a support for **Youtube**, **Vimoe** and **Direct Links**.
- Integrated the **YouTube Data API** for in-room video search.
- Built real-time synchronization for playback controls (play, pause, seek, and playback speed) using **Socket.IO**.
- Developed real-time chat with replies, reactions, typing indicators, message actions, and notifications.
- Implemented room management with invite links and multi-user interactions.

Planets Diseases

A full-stack web application built with **Next.js 16** and **TypeScript** that allows users to browse plant diseases and enables admins to manage structured documentation using **Firebase (Auth & Firestore)** with optional **Redis** caching for performance optimization.

- Developed a rich documentation system using **Editor.js** supporting structured content (headers, tables, images, lists).
- Built responsive UI with **Material-UI (MUI v7)** and managed client state using **TanStack Query v5**.
- Integrated **React Hook Form + Yup** for validation and **Axios** for API communication.

Sports Tournament Scheduling

A distributed microservices system built with **.NET Core**, **Python**, and **React** to manage sports tournament resources and automate optimal timetable generation.

- Developed a Clean **Architecture microservices** backend using ASP.NET Core, utilizing **gRPC** for synchronous inter-service communication and **RabbitMQ** for event-driven orchestration.
- Configured distributed state management and caching using Redis to **queue, track, and persist active generation job metadata** across C# and Python runtimes.
- Integrated Keycloak for Identity and Access Management (**OIDC**) alongside **SpiceDB** to support Zanzibar-inspired fine-grained **relationship authorization**.
- Created a **React/TypeScript** frontend dashboard featuring live updates for active background scheduling pipeline jobs.

ECG Detection System

A **real-time, microservices-based electrocardiogram (ECG)** monitoring and anomaly detection system that ingests high-frequency cardiac signals, runs **AI-driven arrhythmia classifications**, and visualizes live telemetry.

- Engineered a **high-throughput Go** ingestion service utilizing **gRPC streaming** and TimescaleDB bulk-copy protocols to performantly persist high-frequency voltage metrics.
- Architected a decoupled, **event-driven data pipeline using Apache Kafka** to stream heart rate patches to a **Python AI** classifier for real-time arrhythmia detection.
- Implemented a secure **Backend-for-Frontend (BFF)** pattern in ASP.NET Core with **OpenID Connect (OIDC)** and Keycloak to authenticate clients and manage session credentials.
- Developed an interactive React SPA featuring **Apache ECharts** to performantly plot thousands of **timeseries ECG samples** and overlay diagnosed cardiac events.

Automated E-Commerce Deployment Platform

- Refactored an existing monolithic codebase by decoupling it into an independent microservices architecture to improve maintainability and scalability.
- Containerized the separated microservices using Docker and orchestrated the multi-container environment utilizing Docker Compose for seamless inter-service communication.
- Designed and implemented an automated CI/CD pipeline using Jenkins to streamline integration and deployment workflows.
- **Tools:** Jenkins, Docker, Docker Compose, Microservices Architecture